

Individual Development Plan — Sets of Skills

Source: docs/idp/ · Generated 2026-05-11 · 26 Sets of Skills · 130 Skills total

Set 1 **Fundamentals and Theory**

1. 1.1 Own the Team's Data-Systems Theory Foundations
2. 1.2 Own Linux, Shell & Git Operations Practice
3. 1.3 Apply Distributed Systems Theory in Practice
4. 1.4 Adopt Polyglot Programming Literacy
5. 1.5 Apply Data Structures & Algorithms in Production

Set 3 **Storage Layer**

1. 3.1 Architect Layered Storage Practice
2. 3.2 Engineer Partitioning & Clustering Practice
3. 3.3 Evaluate & Operating Table Formats Practice
4. 3.4 Design Columnar Formats & File-Layout Practice
5. 3.5 Select NoSQL, Key-Value & NewSQL Practice
6. 3.6 Architect Real-Time OLAP Solutions Practice
7. 3.7 Architect Specialized Data Stores Practice
8. 3.8 Recognize Hadoop Legacy Ecosystem Practice

Set 4 **SQL and Dimensional Modeling**

1. 4.1 Engineer Advanced SQL Practice
2. 4.2 Design the Team's Dimensional-Modeling Practice (Kimball)
3. 4.3 Evaluate Alternative Modeling Paradigms (Inmon, Data Vault)
4. 4.4 Design Semantic Layer & Metrics Practice
5. 4.5 Operate Federated Query Engines Practice
6. 4.6 Understand Data Visualization & BI Tools (Consumer Perspective for DE)

Set 5 **Orchestration**

1. 5.1 Own the Team's DAG Authoring Practice
2. 5.2 Engineer Orchestrator Resilience & SLA Practice
3. 5.3 Architect Advanced Orchestration Practice (Dynamic, Event-Driven)
4. 5.4 Evaluate Orchestration Alternatives Practice
5. 5.5 Engineer Python for Orchestration Practice

Set 6 **Data Testing Engineering**

1. 6.1 Engineer Generic Test Coverage Practice
2. 6.2 Author Custom Tests & Macros Practice
3. 6.3 Operate Contracts, Great Expectations & Soda Practice
4. 6.4 Engineer Anomaly Detection & DQ KPI Practice

Set 7 **Observability and Operational Reliability**

1. 7.1 Engineer Logs, Metrics & Traces Practice
2. 7.2 Author Operational Metrics & Dashboards Practice
3. 7.3 Define SLI, SLO & Error-Budget Practice
4. 7.4 Lead Incident Management & Postmortem Practice
5. 7.5 Select Observability Stack Ecosystems Practice

Set 8 **Reliability Engineering and Performance**

1. 8.1 Engineer Failure-Handling & Recovery Practice
2. 8.2 Optimize Queries Practice
3. 8.3 Engineer Scaling & Capacity Practice
4. 8.4 Architect HA & Disaster Recovery Practice

Set 9 **Streaming and Event-Driven Architectures**

1. 9.1 Understand Streaming Fundamentals
2. 9.2 Build End-to-End Streaming POC
3. 9.3 Evaluate Batch vs Streaming Decision
4. 9.4 Evaluate Stream Processing Engines (Flink, Spark Streaming, Beam)
5. 9.5 Design Streaming SQL & Stateful Workflows Practice
6. 9.6 Select Messaging Brokers Beyond Kafka

Set 10 **Data Governance**

1. 10.1 Operate Data Catalog & Documentation Practice
2. 10.2 Engineer Column-Level Lineage Practice
3. 10.3 Govern Contracts & Ownership Practice
4. 10.4 Architect Data Products & Data Mesh Practice
5. 10.5 Engineer Feature Stores & API Gateways for Data Products

Set 11 **Security and Compliance**

1. 11.1 Engineer Identity & Access Control Practice
2. 11.2 Operate Secrets Management Practice
3. 11.3 Secure PII & Sensitive Data Practice
4. 11.4 Govern Regulatory Compliance (LGPD/GDPR) Practice

Set 12 **End-to-End Architecture and Systems Thinking**

1. 12.1 Author Architectural Documentation Practice
2. 12.2 Design Systems via Well-Architected Practice
3. 12.3 Apply Systems Thinking Practice
4. 12.4 Architect Multi-Region & Cross-Region Replication Practice
5. 12.5 Drive DataOps Maturity + Strategic Mapping Practice
6. 12.6 Architect Hybrid & Multi-Cloud Strategy Practice

Set 13 **FinOps and Cost Optimization**

1. 13.1 Engineer Cost Visibility Practice
2. 13.2 Optimize Cost Practice
3. 13.3 Define Budgets & Forecasting Practice

Set 14 **CI/CD and Data Workflow**

1. 14.1 Engineer Hooks & Linting Practice
2. 14.2 Engineer Slim CI for dbt Practice
3. 14.3 Define Git Workflow & Release Management Practice
4. 14.4 Select CI/CD Platforms Practice

Set 15 **Infrastructure as Code**

1. 15.1 Engineer Terraform at Scale Practice
2. 15.2 Engineer Kubernetes Manifests & Helm Practice
3. 15.3 Operate GitOps (ArgoCD, Flux) Practice
4. 15.4 Operate Configuration Management Practice

Set 16 **Containerization and Reproducibility**

1. 16.1 Engineer Container Images & Runtimes Practice
2. 16.2 Engineer Dev/Prod Parity Practice
3. 16.3 Architect Kubernetes Workloads Practice

Set 17 **Distributed Big Data Processing**

1. 17.1 Understand Spark Fundamentals Practice
2. 17.2 Operate Spark, Databricks & Dataproc Practice
3. 17.3 Optimize Big Data Workloads Practice

Set 18 **Python for Data Engineering**

1. 18.1 Write Idiomatic Python for Engineering Practice
2. 18.2 Engineer Python Packaging & Dependencies Practice
3. 18.3 Engineer Testing for Data Code Practice
4. 18.4 Optimize Python Performance & Profiling Practice
5. 18.5 Select Canonical Python Libraries for DE Practice
6. 18.6 Engineer Async, Concurrency & Queues Practice
7. 18.7 Architect Python Patterns for DE Practice

Set 19 **AWS for Data Engineering (DEA-C01)**

1. 19.1 Operate AWS Ingestion Services Practice
2. 19.2 Operate AWS Storage Services Practice
3. 19.3 Operate AWS Processing Services Practice
4. 19.4 Operate AWS Orchestration & Operations Practice
5. 19.5 Govern AWS Security & Compliance Practice
6. 19.6 Optimize AWS-Specific Cost + Passing DEA-C01

Set 20 **GCP for Data Engineering**

1. 20.1 Operate GCP Ingestion Services Practice
2. 20.2 Operate GCP Storage Services Practice
3. 20.3 Operate GCP Processing Services Practice
4. 20.4 Operate GCP Orchestration & Operations Practice
5. 20.5 Govern GCP Security & Compliance Practice
6. 20.6 Master BigQuery + Passing PDE

Set 21 **Databricks for Data Engineering**

1. 21.1 Architect Lakehouse with Delta Lake Practice
2. 21.2 Engineer ELT with Spark on Databricks Practice
3. 21.3 Operate Databricks Workspace & Workflows Practice
4. 21.4 Engineer Delta Live Tables & Auto Loader Practice
5. 21.5 Govern Unity Catalog Practice
6. 21.6 Operate Databricks SQL & Serving + Passing Certs

Set 22 **Snowflake for Data Engineering**

1. 22.1 Understand Snowflake Architecture Practice
2. 22.2 Engineer Snowflake Data Loading Practice
3. 22.3 Engineer Streams, Tasks & Dynamic Tables Practice
4. 22.4 Govern Snowflake Security Practice
5. 22.5 Optimize Snowflake Performance & Cost Practice
6. 22.6 Engineer Snowpark, Apps & Data Sharing + Passing Certs

Set 23 **Airflow — Operation and Deep Engineering**

1. 23.1 Understand Airflow Architecture & Internals Practice
2. 23.2 Author Advanced DAGs Practice
3. 23.3 Operate Airflow Providers Ecosystem Practice
4. 23.4 Operate Airflow in Production Practice
5. 23.5 Engineer Testing & CI/CD for DAGs Practice
6. 23.6 Evaluate Orchestrator Alternatives + Passing Astronomer Certs

Set 24 **dbt — Deep Transformation Engineering**

1. 24.1 Engineer dbt Fundamentals & Modeling Practice
2. 24.2 Engineer dbt Testing (Contracts + Unit Tests) Practice
3. 24.3 Author Advanced Macros & Jinja Practice
4. 24.4 Operate dbt Packages & Ecosystem Practice
5. 24.5 Engineer dbt Operations & Deployment Practice
6. 24.6 Architect dbt Cloud, Mesh & Semantic Layer + Passing Cert

Set 25 **Product Vision and Business Orientation**

1. 25.1 Apply Business Understanding Practice
2. 25.2 Drive Scope & Priority Negotiation Practice
3. 25.3 Drive Simplification & MVP Thinking Practice
4. 25.4 Design Experimentation & Data-Driven Decisions Practice

Set 26 **Technical Communication and Influence**

1. 26.1 Author Technical Documentation as Code Practice
2. 26.2 Communicate to Non-Technical Audiences Practice
3. 26.3 Drive Thought Leadership Practice

Set 27 **Mentorship, Leadership, and Technical Responsibility**

1. 27.1 Author Technical Standards Practice
2. 27.2 Lead Technical Initiatives Practice
3. 27.3 Own End-to-End Technical Responsibility Practice
4. 27.4 Lead Mentorship, Pairing & Code Review Practice
5. 27.5 Drive Organization-Wide Standardization Practice